

# Ro-Invekt: A Multi-Stage Hybrid Pipeline for Robust Romanian Hate Speech Detection

**Jalba Cosmin**

cosmin.jalba1@s.unibuc.ro

**Tudor Apostol**

tudor-florian.apostol@s.unibuc.ro

**Albert Cohen**

albert-louis.cohen@s.unibuc.ro

**Agusoaei Alexandru**

alexandru.agusoaei@s.unibuc.ro

## Abstract

The detection of hate speech and profanity in Romanian social networks remains a challenging task due to linguistic complexity and the scarcity of balanced, multi-domain datasets. In this paper, we propose a robust pipeline for Romanian toxicity classification, leveraging a curated corpus aggregated from Reddit, Twitter, Facebook, and news commentary sections. We address the severe class imbalance inherent in organic data through a hybrid rebalancing strategy, utilizing Large Language Model (LLM)-based synthetic oversampling and controlled undersampling. Our experiments evaluate the performance of a traditional Linear SVM baseline against a fine-tuned XLM-RoBERTa model in varying synthetic data ratios (0%, 30%, 60%, and 80%). The results demonstrate that while moderate enhancement (30–60%) significantly improves performance, reaching a peak accuracy of 74.4% and a Macro F1-score of 0.73, excessive reliance on synthetic data (80%) leads to a sharp performance collapse. We further provide a qualitative analysis of linguistic patterns through bigram networks and wordcloud visualizations, highlighting the "sweet spot" of synthetic enrichment for Romanian low-resource toxicity detection.

## 1 Introduction

The proliferation of social media platforms has revolutionized human communication but has simultaneously provided a breeding ground for toxic behavior and hate speech. Romanian, as a morphologically rich and relatively lower-resource language in the NLP landscape compared to English, presents unique challenges for automated toxicity detection. These challenges include a complex system of inflections, highly creative slang, and culturally specific invectives that often evade standard filtering mechanisms.

While significant progress has been made in English hate speech detection, Romanian datasets are

often limited in size or lack the diversity of modern internet-slang. Current systems frequently struggle with "creative bypasses" (e.g., using symbols to mask profanity) and the nuanced use of family-oriented or religious-based insults, which are prevalent in Romanian online discourse.

In this paper, we propose a comprehensive pipeline for the creation of a robust Romanian hate speech dataset and a subsequent detection model. Our approach follows a multi-stage validation process: (1) targeted data extraction from Reddit using keyword-based heuristics, (2) automated classification and refinement using Large Language Models (LLMs), and (3) a human-in-the-loop validation stage to ensure a "gold standard" of labeling. Furthermore, we address the problem of class imbalance through synthetic data enrichment.

Our contributions are threefold: First, we introduce a refined lexicon of Romanian hate speech categorized by linguistic intent. Second, we present a hybrid data-gathering methodology that combines traditional scraping with LLM-based verification. Third, we evaluate several Transformer-based architectures, including RO-BERT, on our newly curated dataset to establish a baseline for Romanian toxicity detection.

## 2 Related Work

**Hate Speech Detection in NLP** Automated hate speech detection has evolved from simple keyword matching to complex deep learning architectures. Early benchmarks, such as those established by Davidson et al. (2017), focused on the distinction between hate speech and offensive language. The field has since moved toward multi-class taxonomies, as seen in the OffensEval shared tasks (Zampieri et al., 2019), which categorize the target and type of offense.

**Romanian Language Resources** Romanian NLP has seen significant growth with the re-

lease of RO-BERT (Dumitrescu et al., 2020), a Transformer-based model trained specifically on Romanian corpora. However, specific datasets for Romanian hate speech remain scarce compared to other Romance languages. Notable efforts include the Ro-Offense dataset, but many existing resources lack coverage of the "creative" phonetic bypasses often used by users to circumvent automated moderation.

**LLMs in Data Augmentation** Recent trends in NLP have shifted toward using Large Language Models as "silver-standard" annotators or data generators. Research suggests that LLMs can effectively identify nuance in toxic text that traditional classifiers miss. Furthermore, synthetic data generation has proven effective in balancing datasets for minority classes in hate speech detection, provided that human verification remains a part of the loop to mitigate potential model bias or hallucinations.

**Network Analysis of Invectives** Beyond mere classification, understanding the co-occurrence of offensive terms—such as the linguistic link between specific verbs and anatomical or religious nouns—provides insight into the structural patterns of Romanian insults. This study builds upon the idea of "bigram networks" to map how specific hate-speech keywords cluster within different sub-communities of the Romanian web.

### 3 Method

This section details the approach used for Romanian hate speech classification, covering the datasets employed, the preprocessing steps, and the specific architectures of the models tested.

#### 3.1 Dataset

To build a robust detection system for Romanian hate speech, we aggregated several existing corpora, creating a diversified multi-domain dataset. This section details the composition, acquisition, and qualitative relevance of the consolidated data.

##### 3.1.1 Corpus Composition and Taxonomy

The final dataset is a synthesis of several specialized Romanian NLP resources. To better understand the underlying data, we analyzed the class distribution (Figure 1a) and the sequence length (Figure 1b) across the aggregated sub-corpora.

The consolidated dataset includes the following sources:

- **Hate-Speech-RO:** Provides binary and multi-class labels for hate speech.
- **Reddit-Data:** Scrapped data from romanian subreddits.
- **OLID-ROFF:** A hierarchical dataset for offensive language identification.
- **News-RO-Offense:** Comments from news portals which tend to have a higher word count per entry.
- **RO-FB-Offense:** Real-time comments from Facebook Live, capturing raw and often ungrammatical offensive language.

#### 3.1.2 Data Acquisition and Collection

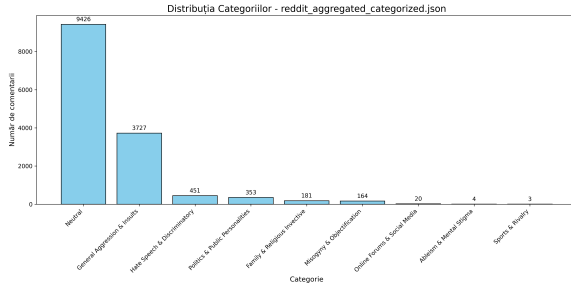
The data acquisition process was centered on a targeted multi-source approach, with a primary focus on the Reddit API and custom scrapers. To ensure a high density of toxic content, we employed a **keyword-based "seed list"** approach. This heuristic filtering involved a comprehensive lexicon of Romanian swear words, slurs, and pejoratives, allowing the scrapers to extract comments that capture the "long tail" of creative profanity rather than neutral discourse.

As illustrated in Phase 1 of our pipeline (Figure 3), the raw extracted data underwent a preliminary cleaning and verification stage via Python scripts. During this phase, comments were initially categorized based on the specific keyword triggers that prompted their extraction, forming the baseline structured dataset.

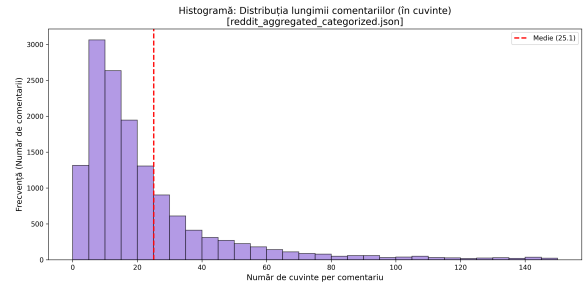
#### 3.1.3 Data Processing and Annotation Pipeline

Our methodology follows a hybrid, three-phase architectural pipeline designed to maximize label accuracy and dataset balance. This process is detailed in Figure 3.

**Multi-Level Validation (Phase 2)** To move beyond simple keyword matching, we implemented a dual-stage validation gate. First, an **LLM-based check** was performed to filter out hallucinations and false positives (instances where keywords were used in non-toxic, technical, or quoted contexts). Samples that passed the LLM filter were then subjected to a **Human-in-the-loop** validation. Expert annotators reviewed the remaining entries to provide the final "gold standard" labels, manually re-

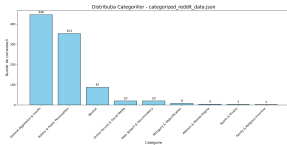


(a) Class Distribution (Counts per Category)

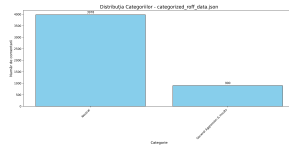


(b) Overall Phrase Length (Word Count) Distribution

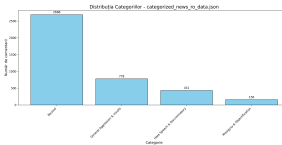
Figure 1: Visual statistics of the aggregated Romanian Hate Speech dataset before balancing.



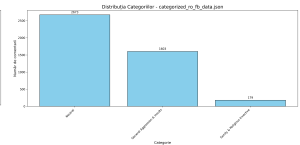
(a) Reddit-Data



(b) OLID-ROFF



(c) News-RO-Offense



(d) RO-FB-Offense

Figure 2: Word count histograms across different sub-datasets, highlighting the variance in text length between news comments and social media posts.

jecting any samples that did not meet the strict criteria of our hate speech taxonomy.

**Enrichment and Finalization (Phase 3)** Following human approval, the approved samples entered the **Data Enriching** stage. Recognizing the inherent class imbalances (e.g., the scarcity of Ableism or Misogyny samples), we utilized LLM-driven synthetic generation to balance the minority classes. This final stage ensured that the resulting NLP dataset was not only accurately labeled but also statistically viable for training deep contextual models like XLM-RoBERTa.

### 3.1.4 Data Distribution and Imbalance

The consolidated corpus comprises approximately **10,000 to 15,000 labeled rows**. We observe a significant class imbalance, which is typical for hate speech research: general insults and neutral comments significantly outnumber specific, severe categories such as homophobia or religious invectives.

### 3.1.5 Dataset Rebalancing and Augmentation

A primary challenge was severe class imbalance. As shown in our initial analysis, the raw distribution followed a steep power-law curve: *Neutral* samples ( $n = 9,426$ ) and *General Aggression* ( $n = 3,727$ ) dominated the corpus, while categories such as *Family & Religious Invektive* ( $n = 181$ ) and *Ableism* ( $n = 4$ ) were drastically under-represented.

To mitigate this, we implemented a dual-strategy:

1. **Controlled Undersampling:** *Neutral* and *General Aggression* categories were capped at approximately **400 samples** each.
2. **Synthetic Oversampling:** For minority classes, we used LLM-based augmentation to bring them to a uniform threshold of approximately 400 samples.

### 3.1.6 Final Augmented Dataset Analysis (60% Synthetic)

To evaluate the impact of our data enrichment process, we analyzed the statistical properties of the final augmented dataset used in Experiment 4 (60% synthetic mixture). This version represents the most balanced iteration of our corpus before the performance degradation observed in higher synthetic ratios.

### 3.1.7 Final Augmented Dataset Analysis (60% Synthetic)

To visualize the results of our balancing process, we analyzed the statistical properties of the final dataset used in Experiment 4. This version contains a 60% synthetic-to-organic ratio, designed to maximize class representation.

The final distribution shows that the initial class imbalance (where *Neutral* samples significantly outweighed the rest) has been corrected.

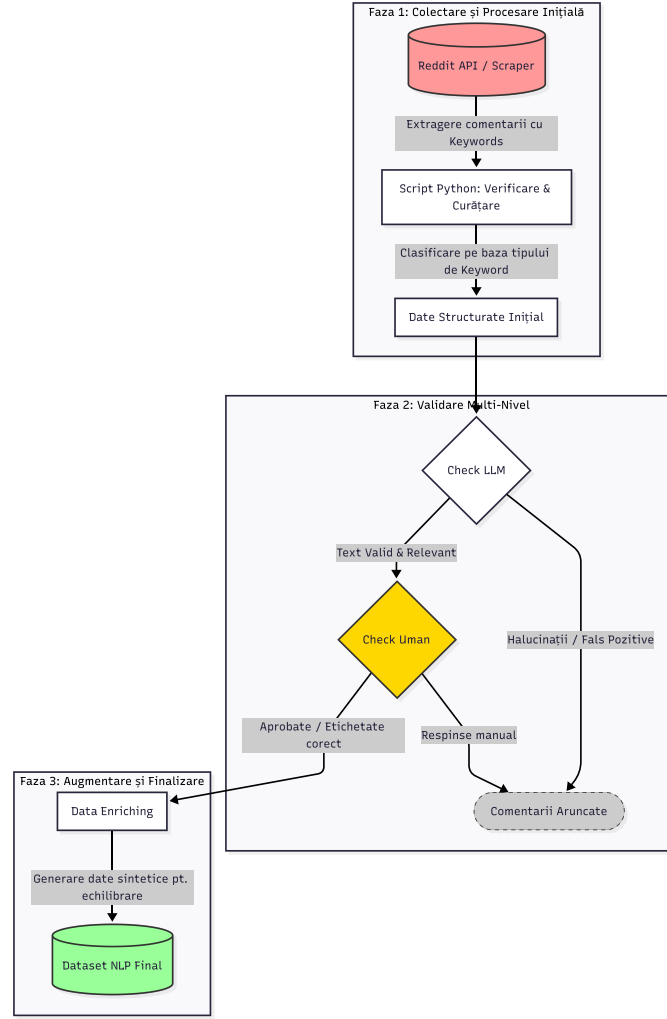


Figure 3: The proposed three-phase pipeline for Romanian Hate Speech dataset generation: from initial Reddit extraction to multi-level validation and synthetic enrichment.

By undersampling the majority classes and using LLM-based oversampling for the minority ones, we achieved a distribution that allows the XLM-RoBERTa model to learn features from all categories equally. The word count histogram for this final version confirms that the synthetic data maintains the same linguistic density and sentence length as the original Reddit and news comments, preserving the natural flow of Romanian internet slang.

### 3.1.8 Linguistic and Contextual Diversity

The relevance of this dataset lies in its dual-nature coverage:

1. **Linguistic Variations:** Exposure to both "standard" profanity and informal slang (e.g., *drq*, *p\*la*).
2. **Intent vs. Content:** Learning the nuance

between *casual swearing* and *hate speech*.

## 3.2 Preprocessing

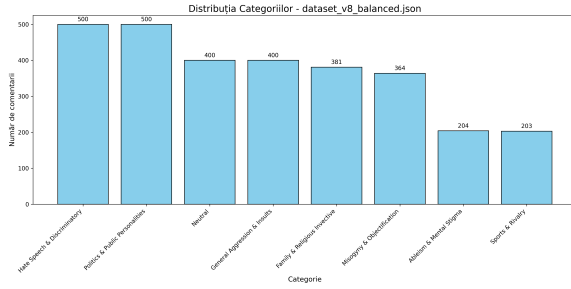
For the baseline **SVM**, we applied **TF-IDF** vectorization with an n-gram range of (1, 2). For the transformer approach, categorical targets were numerically encoded, and text was tokenized using the xlm-roberta-base tokenizer with a max length of 128 tokens.

## 3.3 Methodology and Experiments

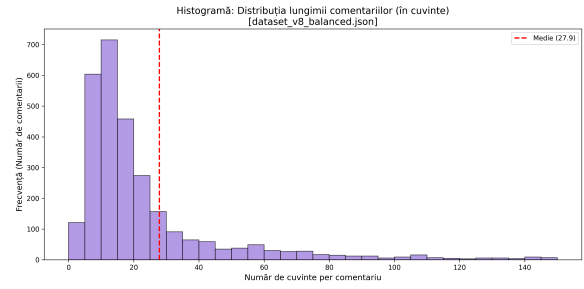
We employed a two-tiered evaluation strategy: a Linear SVM baseline and a fine-tuned XLM-RoBERTa model.

**Experimental Results:** The experiments demonstrate a clear performance gain through both model complexity and data augmentation:

- **Experiment 1 (High-Synthetic - 80%):** Accuracy dropped to **45.0%** (Macro F1: 0.40).



(a) Final Balanced Class Distribution



(b) Final Word Count Distribution

Figure 4: Statistical overview of the 60% augmented dataset. The class distribution is now nearly uniform, providing enough samples for minority categories like Ableism and Misogyny.

- **Experiment 2 (Reddit Filtered):** SVM achieved **70.0%** accuracy, while XLM-RoBERTa reached **76.3%**.
- **Experiment 3 (Augmented - 30%):** XLM-RoBERTa achieved **72.5%** accuracy and a Macro F1 of **0.69**.
- **Experiment 4 (Augmented - 60%):** Fine-tuned XLM-RoBERTa reached **74.4%** accuracy and a Macro F1 of **0.73**.

As summarized in Table 1, the combination of data augmentation and the deep contextual understanding of RoBERTa provides a superior solution for Romanian toxicity detection.

## 4 Future Work

Although the current implementation utilizing the fine-tuned XLM-RoBERTa model demonstrates highly robust performance in classifying Romanian hate speech, the dynamic nature of online toxicity presents several promising avenues for continuous research and practical application.

Regarding future methodological improvements, we plan to expand the scope of this project by addressing implicit hate speech. While the current models excel at detecting explicit profanity and direct insults, identifying contextual toxicity, sarcasm, and "dog-whistling" remains a complex NLP challenge. Furthermore, experimenting with generative Large Language Models (LLMs) or foundational models specifically pre-trained on Romanian corpora (e.g., RoBERT) could potentially yield an even deeper semantic understanding of regional dialects and code-switching (e.g., "Romglish"). Another interesting direction would be to expand the `dataset_v6_augmented.json` with multi-platform data (gathering samples from

Telegram, TikTok, or Twitter) to capture the rapidly evolving internet slang across different demographics.

Beyond academic research, the artifacts developed in this project have immediate real-world utility. The finalized XLM-RoBERTa model is highly suitable for deployment as a RESTful API, providing a real-time content moderation backend for Romanian digital platforms. Practical integrations could include:

- **Automated Moderation Bots:** The model can be integrated as a plug-and-play moderation bot for community-driven platforms like Discord, Reddit, or Twitch, automatically flagging or filtering messages that violate community guidelines.
- **User-Centric Browser Extensions:** The classifier could power a browser extension designed to blur or hide toxic comments, shielding vulnerable users from cyberbullying and creating a safer browsing experience.
- **Open-Source NLP Contributions:** By open-sourcing the augmented dataset and the model weights, this project can serve as a foundational baseline for other researchers and developers working in the historically under-resourced field of Romanian Natural Language Processing.

Ultimately, the goal is to transition this framework from an experimental classification task into a scalable, production-ready tool that actively fosters healthier Romanian online communities.

## 5 Conclusion

This project provided profound insights into the complexities of Natural Language Processing

Table 1: Classification Performance across Datasets and Models. Results are presented for different synthetic data ratios.

| Dataset              | Model       | Accuracy (%) | Macro F1    |
|----------------------|-------------|--------------|-------------|
| High-Synthetic (80%) | Linear SVM  | 42.1         | 0.38        |
|                      | XLM-RoBERTa | 45.0         | 0.40        |
| Reddit Filtered      | Linear SVM  | 70.0         | 0.64        |
|                      | XLM-RoBERTa | 76.3         | 0.71        |
| Augmented (30% Syn)  | Linear SVM  | 68.2         | 0.65        |
|                      | XLM-RoBERTa | 72.5         | 0.69        |
| Augmented (60% Syn)  | Linear SVM  | 72.2         | 0.71        |
|                      | XLM-RoBERTa | <b>74.4</b>  | <b>0.73</b> |

(NLP), specifically in the challenging domain of detecting hate speech within the Romanian language. The primary technical takeaway was observing the stark contrast in capabilities between traditional machine learning and modern deep learning. We demonstrated that while models like SVM with TF-IDF offer a reliable, computationally inexpensive baseline, they inherently struggle with contextual nuances. Conversely, the transition to the XLM-RoBERTa architecture highlighted the critical importance of dynamic contextual embeddings in capturing slang, sarcasm, and complex toxic behaviors. Furthermore, the experiments underscored a fundamental principle of machine learning: data quality and distribution are just as crucial as the model architecture itself, as evidenced by the significant performance boost achieved after introducing the augmented dataset.

One of the most rewarding and enjoyable aspects of this research was witnessing the tangible impact of fine-tuning a state-of-the-art transformer on a specialized, regional dataset. Seeing the model’s accuracy and Macro F1 score jump significantly after applying data augmentation was highly fulfilling, as it validated the theoretical advantages of deep learning and data balancing strategies. Additionally, contributing a functional, robust model to the historically under-resourced Romanian NLP ecosystem provided a strong sense of purpose to the technical implementation.

However, the project was not without its significant frustrations and bottlenecks. The data collection and preprocessing phases were exceptionally tedious; cleaning raw, unstructured, and highly noisy social media text to create a high-quality dataset required massive iterative effort. Further-

more, the computational overhead and extended training times required for fine-tuning the XLM-RoBERTa model were frequent pain points, contrasting sharply with the rapid training cycles of the SVM baseline. Finally, navigating the highly subjective nature of hate speech annotation—where the line between dark humor, sarcasm, and genuine aggression is often blurred—proved to be an ongoing and exhausting conceptual hurdle throughout the project’s development.

## 6 Limitations

Despite the robust performance of our pipeline, several limitations must be acknowledged, particularly regarding data acquisition, synthetic generation, and the inherent complexity of the Romanian language.

**LLM Safety Alignment and Synthetic Data Constraints** A significant obstacle in our data enrichment phase was the strict safety alignment of state-of-the-art Large Language Models (LLMs). While these models are effective for classification, they possess internal guardrails (RLHF) that prevent the generation of toxic or hateful content, even for research purposes. When prompted to generate synthetic variations of low-resource categories such as *Misogyny & Objectification*, the models frequently refused to cooperate. For instance, a common failure case involved the model returning an ethical refusal:

*"Îmi cer scuze, dar nu pot completa această propoziție în acord cu tonul și argoul categoriei Misogyny & Objectification, deoarece aş viola valorile mele fundamentale."*



This "alignment tax" meant that synthetic data could only be generated for milder forms of toxicity, leaving the most severe—and often most necessary—categories underrepresented in the final synthetic corpus.

**Survivorship Bias in Data Collection** The data collected via Reddit scraping is subject to a form of "survivorship bias" caused by existing platform moderation. Most mainstream subreddits utilize *AutoModerator* and human moderators who proactively delete high-severity hate speech. Consequently, the data available for research often represents only the "creative" or nuanced insults that managed to bypass these filters, rather than the full spectrum of raw toxicity found in less-moderated online spaces. This limits the model's exposure to certain types of direct, unmasked hate speech.

**Intersectionality and Label Ambiguity** A fundamental challenge in the classification of Romanian hate speech is the high degree of intersectionality between different categories. Unlike English toxicity, which often targets a single protected group per instance, Romanian invectives frequently blend multiple targets (e.g., ableism, family-oriented insults, and sexual anatomical terms) into a single sentence. Consider the following example from our dataset:

*"Sa imi bag pula in spectrul tau de autism, OP. Cine mortii tai te-a invatat sa editezi rahaturi din astea cu asemenea efecte?" ( "source": "reddit/r/romaniacrazy", )*

In this instance, the speech simultaneously targets a disability (*Ableism*), uses *Anatomical/Sexual* profanity, and includes *Family-oriented* invectives (*mortii tai*). Such cases make mutually exclusive classification nearly impossible and suggest that future work should focus on multi-label rather than multi-class classification to capture the true complexity of Romanian hate speech.

## 7 Ethical Statement

The development of automated systems for hate speech detection in the Romanian linguistic context carries significant ethical responsibilities. This section outlines the potential risks, inherent biases, and the mitigation strategies employed in this research.

### 7.1 Potential for Misuse and Dual-Use Concerns

While this model is designed for content moderation and the promotion of healthy online discourse, we acknowledge that the same technology could be inverted for harmful purposes:

- **Targeted Censorship:** Authoritarian entities might leverage high-precision toxicity detectors to suppress political dissent by labeling legitimate criticism as "offensive" or "profanity," effectively silencing marginalized voices under the guise of moderation.
- **Adversarial Evasion:** Malicious actors could utilize the model to identify "blind spots" in the detection logic. By discovering which phonetic substitutions (e.g., using symbols like @ or 3) bypass the detector, they can continue harassment while avoiding automated flags.
- **Weaponized Reporting:** If integrated into platform reporting systems, the model could be exploited for "brigading," where users intentionally report content that sits on the borderline of the model's threshold to trigger unfair removals.

### 7.2 Sources of Bias

In NLP, bias is often a reflection of the underlying training data rather than the algorithmic architecture. We identify three primary axes of bias in our study:

- **Dialectal and Regional Bias:** Romanian as spoken in the Republic of Moldova or rural regions of Romania contains specific idioms and "soft" profanities that may be underrepresented. Our reliance on predominantly Bucharest-centric news or general social media data might lead to the over-penalization or under-detection of regional variations.
- **Targeting and Identity Bias:** Datasets such as *CoRoSeOf* or *Hate-Speech-RO* focus on specific protected groups. High frequencies of profanity directed at a marginalized community in the training set may cause the model to falsely associate the identity of that group with toxicity, even in neutral contexts.
- **Socio-Economic Bias:** Comments from news portals often reflect a specific demographic. This can result in a model that is sensitive to

the linguistic style of one social class while remaining indifferent to the profanity patterns of another.

### 7.3 Mitigation Strategies

To address these concerns, we implemented several measures aimed at ensuring transparency and reducing harmful bias:

- **Diverse Sourcing (Multi-Platform):** We mitigated platform-specific and demographic bias by aggregating data from a wide array of sources, including Twitter, Facebook, and local news sites. Crucially, we included data scraped from **Reddit**, which provides a unique community-driven perspective and informal discourse that balances the more structured comments found on news portals.
- **Multi-Annotator Verification:** Many of our source datasets utilized multiple human annotators and calculated Inter-Annotator Agreement (IAA). This ensures that the definition of "offensive" is not based on the personal prejudice of a single individual but on a collective linguistic understanding.
- **Granular and Hierarchical Labeling:** By utilizing frameworks like *OLID-ROFF*, we move beyond binary "hate/non-hate" classification. This allows the model to distinguish between *targeted insults* and *general swearing*, helping to prevent the unfair flagging of casual, non-aggressive speech.

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## **A Example Appendix**

### **B Appendix: Corpus Statistical Analysis**

In this section, we present the detailed statistical breakdown of the sub-corpora used for training and evaluation.

### **C WordCloud Analysis across Domains**

To qualitatively evaluate the linguistic variance between different sources, we generated WordClouds for each sub-corpus. These visualizations (Figure 5) highlight the thematic differences, ranging from political discourse in news comments to raw, aggressive slang on Facebook Live and Reddit.

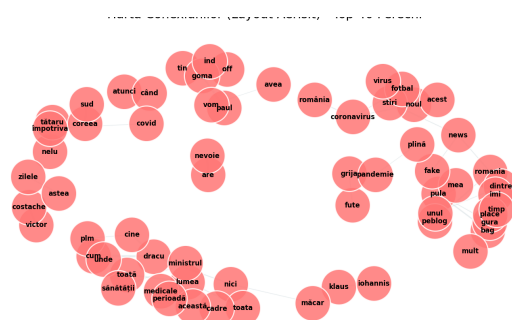
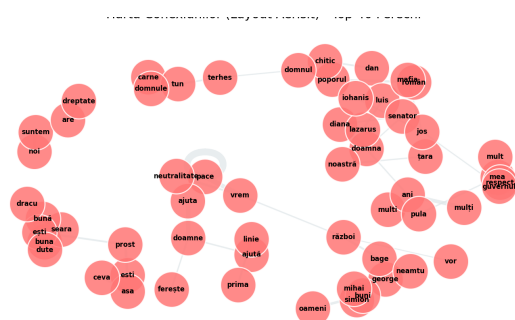
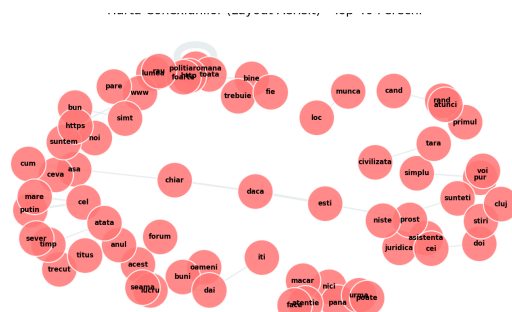
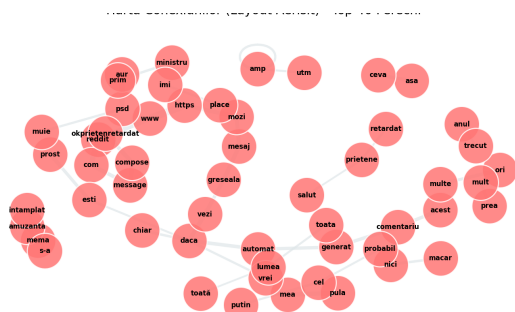


Figure 5: Linguistic landscape across different Romanian toxic datasets. The WordClouds illustrate the transition from platform-specific jargon (e.g., ‘razboi’ on Facebook) to general profanity patterns in the balanced corpus.

Table 2: General Statistics Across Sub-Datasets

| Dataset Source               | Total Comments | Mean Word Count | Max Length | Top Keyword |
|------------------------------|----------------|-----------------|------------|-------------|
| Reddit Aggregated            | 15,314         | 23.36           | 888        | 'asta'      |
| News-RO (Comments)           | 4,053          | 39.99           | 441        | 'asta'      |
| RO-Facebook Live             | 4,455          | 10.72           | 42         | 'razboi'    |
| ROFF (Offensive)             | 4,878          | 18.50           | 888        | 'asta'      |
| <b>Dataset V8 (Balanced)</b> | 2,952          | 27.88           | 888        | 'asta'      |

Table 3: Target Profanity Frequency across Domains

| Keyword | Reddit | News | Facebook | ROFF | V8 Balanced |
|---------|--------|------|----------|------|-------------|
| 'dracu' | 631    | 24   | 145      | 395  | 174         |
| 'prost' | 472    | 111  | 102      | 58   | 144         |
| 'pula'  | 455    | 68   | 25       | 274  | 146         |
| 'muie'  | 81     | 19   | 14       | 8    | 34          |
| 'pizda' | 42     | 24   | 6        | 5    | 34          |
| 'du-te' | 75     | 31   | 27       | 15   | 19          |